

Automato com Pilha - Introdução à Teoria da Computação

4. Construa um APD.

* $\{a^m b^n c^{m+n} \mid m, n \geq 0\}$

$G = (\{S, B\}, \{a, b, c\}, (P, S))$

$P = \{S \rightarrow aS \mid B\}$

$B \rightarrow bB \mid \epsilon$

(X Y Z)
↓ ↓ ↓
I LP EP
L.a L.d R.a

B C

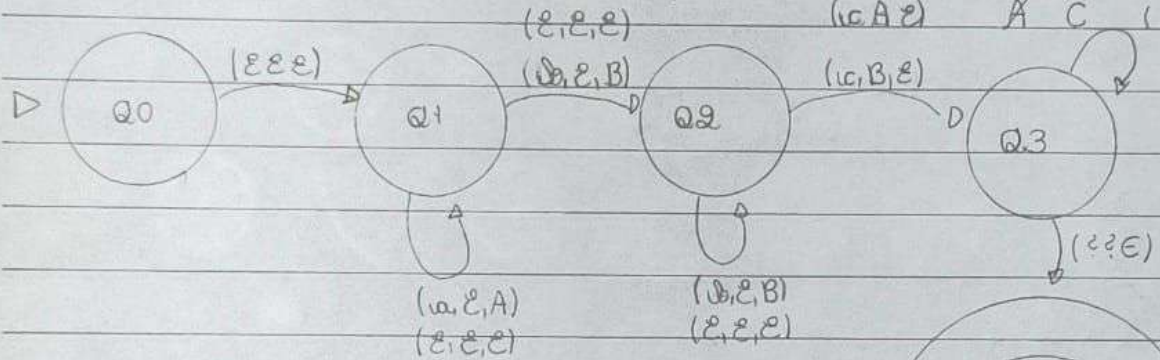
B

B

A C (ε, ε, ε)

(ε, ε, ε) A C (a, A, ε)

(a, A, ε) A C (a, B, ε)



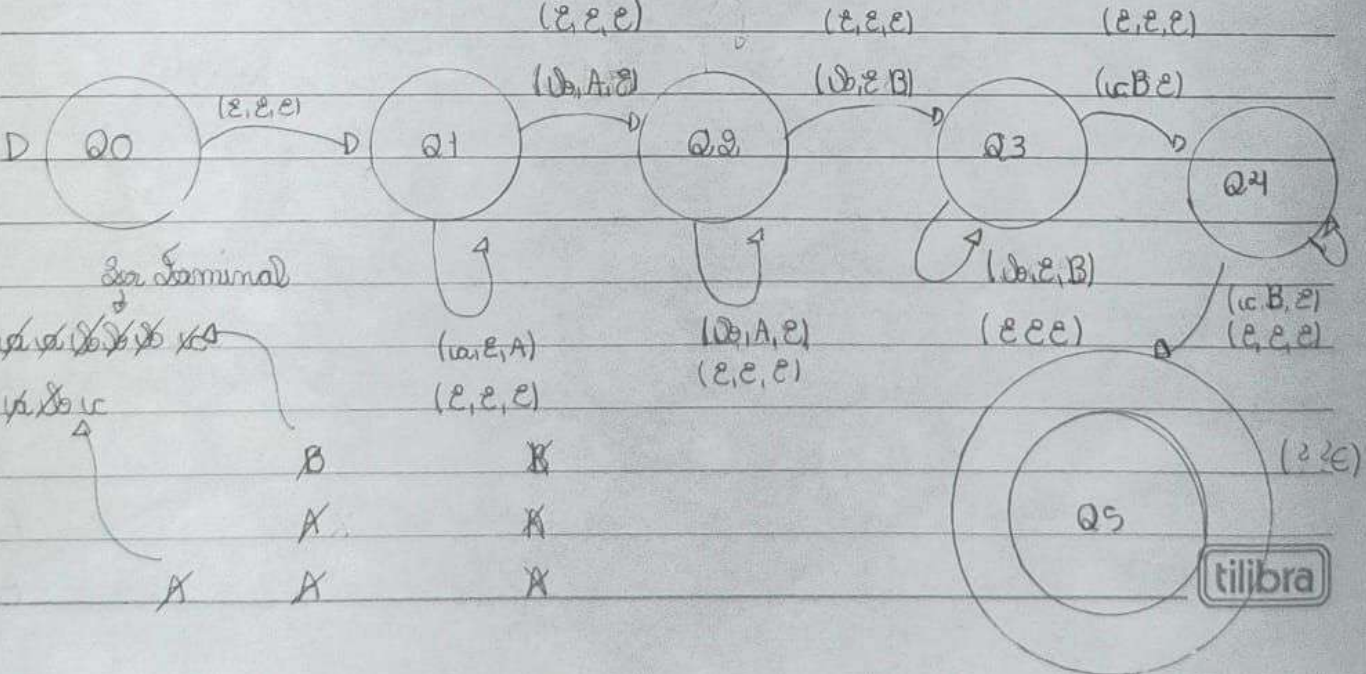
~~✓~~ ~~✓~~ ~~✓~~ ~~✓~~ ~~✓~~ ~~✓~~

B

A

A

* $\{a^m b^n c^{m+n} \mid m, n \geq 0\}$



Seja eliminado

✓ ~~✓~~ ~~✓~~ ~~✓~~ ~~✓~~ ~~✓~~

x ~~✓~~ ~~✓~~ ~~✓~~ ~~✓~~

B

B

A

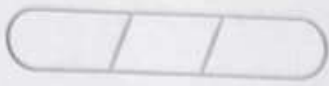
A

A

A

A

tilibra



(X, Y, Z)
 $\downarrow \downarrow \downarrow$ Empfänger
 T, LP, EP
 Lia, Lo, Rho

* $\{1^n \mid 1 \leq n \leq 3\}$

1a 1b

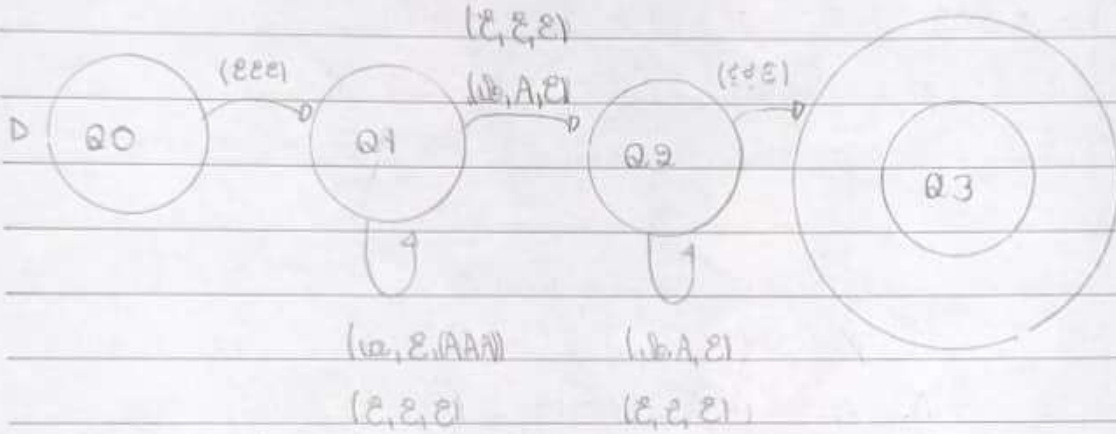
1a 1b 1c 1d

1a 1b 1c

1a 1b 1c 1d

1a 1b 1c 1d

1a 1b 1c 1d

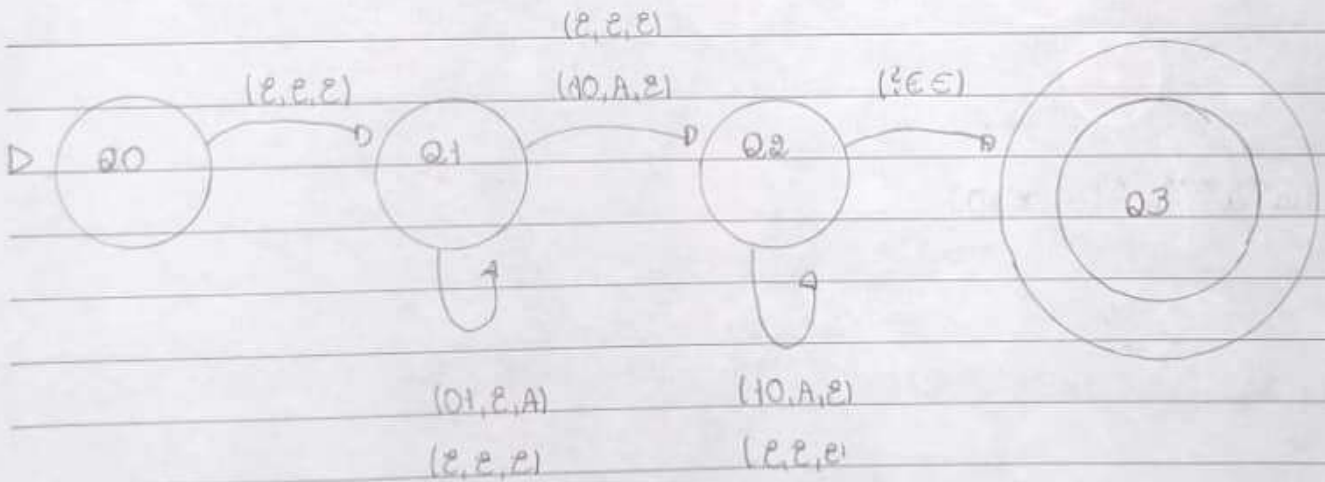


✓ 1a 1b 1c 1d

✗ 1a 1b 1c 1d

✓ 1a 1b 1c 1d

* $\{1^n \mid 1 \leq n \leq 3\}$

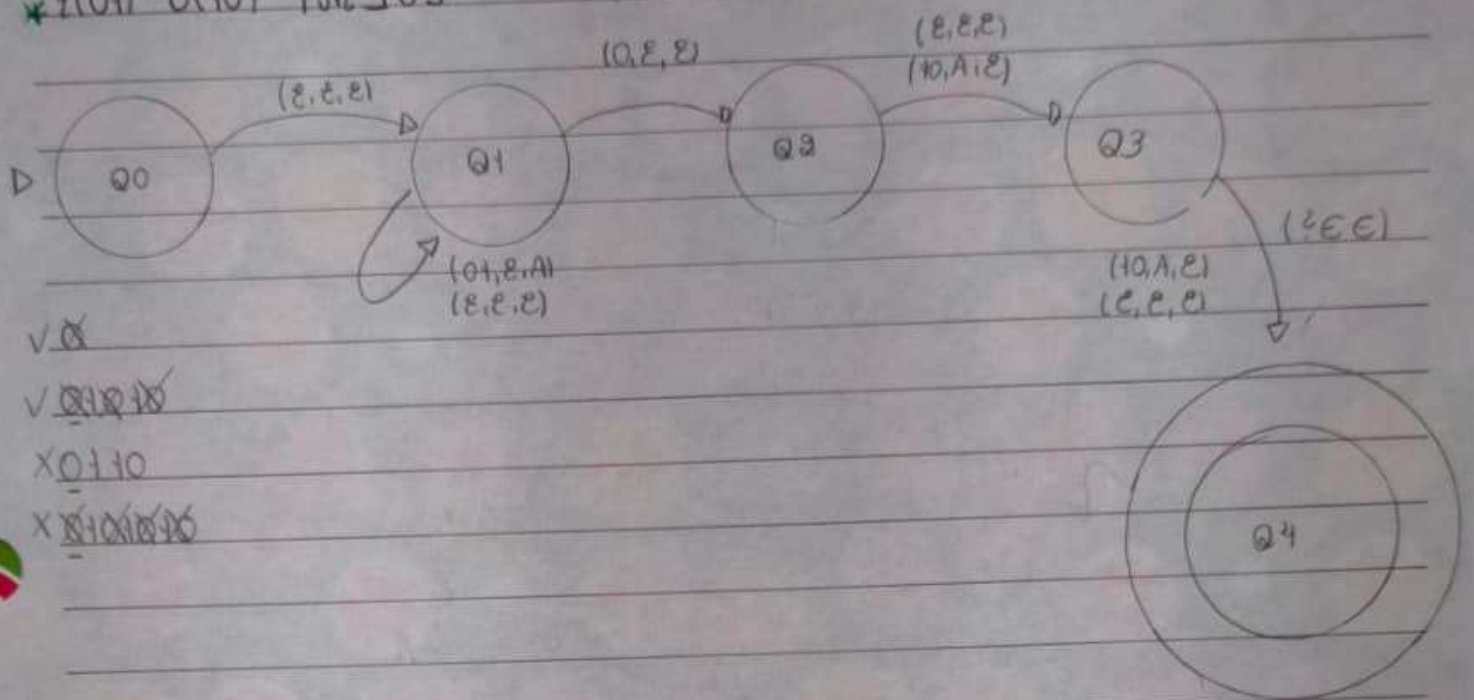


✓ 1a 1b

✓ 1a 1b 1c

✗ 1a 1b

* $1(01)^m 0(10)^m 1$ m ≥ 0



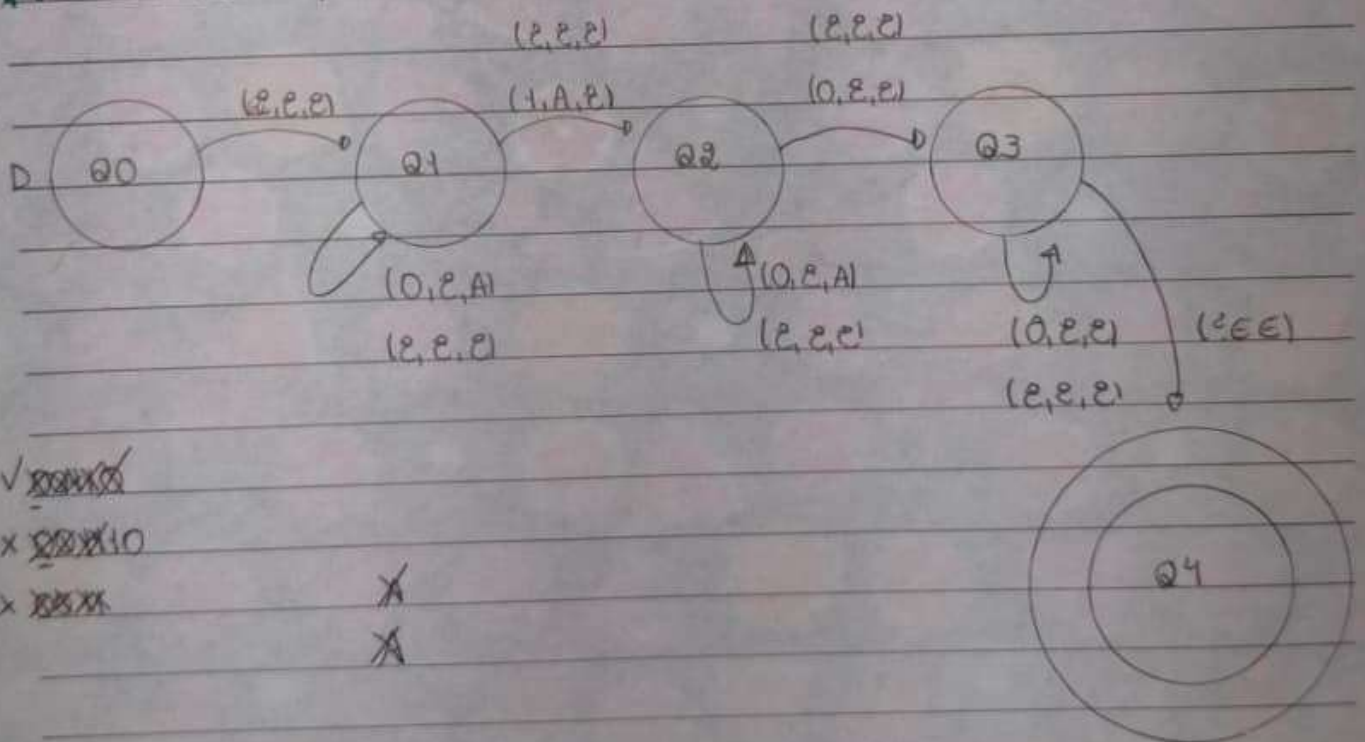
✓ ~~0~~

✓ ~~01010~~

✗ ~~0110~~

✗ ~~0101010~~

* $20^m 1^m 0^m 1$ m ≥ 0 , $n \geq 1$



✓ ~~00000~~

✗ ~~000010~~

✗ ~~00000~~

✗

✗